Fault Code: 119

Fuel Rail Pressure Sensor Circuit - Out of Range Low

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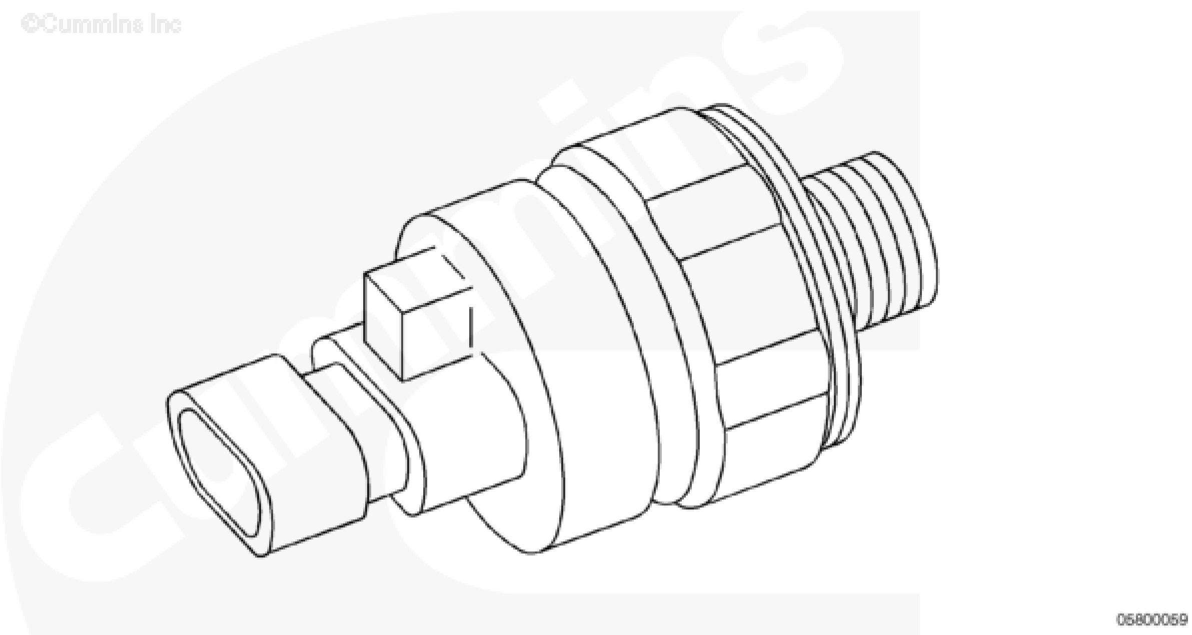
Overview

Codes	Reason	Effect
Fault Code: 119 PID(P): SPN: FMI: Lamp: Red SRT:	Fuel Rail Pressure Sensor Circuit - Out of Range Low. The fuel rail pressure is low.	The Centinel™ system will **not** function.



Circuit Description

The fuel rail pressure sensor monitors the fuel rail pressure. When the sensor fails high or low, the fuel rail pressure sensor fault will occur.

Component Location

The fuel rail pressure sensor is located on the fuel connecting block on the Centinel™ oil control valve mounting bracket.

Warnings and Cautions

⚠ CAUTION ⚠

To avoid pin and harness damage, use the following test leads when taking measurements:
Male Cannon, Metri-Pack, Deutsch test lead, Part Number 3822758
Female AMP, Metri-Pack, Deutsch test lead, Part Number 3822917
Male Deutsch test lead, Part Number 3823993
Female Deutsch test lead, Part Number 3823994.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fuel rail pressure sensor.	
	STEP 1A. Inspect the fuel rail pressure sensor and Centinel™ harness connectors.	No damaged pins
	STEP 1B. Check the Centinel™ control module fuel rail pressure sensor supply voltage.	4.75 to 5.25 VDC
	STEP 1C. Check the fuel rail pressure sensor for an open.	Less than 2M ohms
STEP 2.	Check the Centinel™ harness.	
	STEP 2A. Inspect the Centinel™ harness and the Centinel™ control module connectors.	No damaged pins
	STEP 2B. Check for an open circuit.	Less than 10 ohms
	STEP 2C. Check for a short circuit to ground.	More than 1k ohms
	STEP 2D. Check for a short circuit from pin to pin.	More than 1k ohms
STEP 3.	Clear the fault codes.	
	STEP 3A. Disable the fault code.	Fault Code 119 inactive

STEP 1. Check the fuel rail pressure sensor.

STEP 1A. Inspect the fuel rail pressure sensor and the Centinel™ harness connector.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the Centinel™ harness from the fuel rail pressure sensor.

<table>
<tr><th>Action</th><th>Specification/Repair</th><th>Next Step</th></tr>
<tr><td rowspan="2">Loose connector
Corroded pins
Bent or broken pins
Pushed back or expanded pins
Moisture in or on the connector
Missing or damaged connector seals
Dirt or debris in or on the connector pins
Wire insulation damage
Connector shell broken
Damaged locking tab connector.
For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).</td><td>No damaged pins</td><td>1B</td></tr>
<tr><td>Repair the damaged pins
Repair or replace the Centinel™ harness or the fuel rail pressure sensor, whichever has damaged pins.
Repair the Centinel™ harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html).
Replace the Centinel™ harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).
Replace the fuel rail pressure sensor. Refer to Procedure 019-115 (/qs3/pubsys2/xml/en/procedures/96/96-019-115.html).
Install the appropriate connector seal if damaged or missing.</td><td>3A</td></tr>
</table>

STEP 1B. Check the Centinel™ control module fuel rail pressure sensor supply voltage.

Conditions :

- Turn the keyswitch ON.
- Disconnect the fuel rail pressure sensor from the Centinel™ harness.

Action	Specification/Repair	Next Step
• Measure the supply voltage from pin A to pin B on the harness side of the fuel rail pressure sensor. Refer to the wiring diagram for connector pin identification.	4.75 to 5.25 VDC	**1C**
	Replace the harness or Centinel™ control module Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	**3A**

STEP 1C. Check the fuel rail pressure sensor for an open.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the fuel rail pressure sensor from the Centinel™ harness.

Action	Specification/Repair	Next Step
• Measure the resistance from pin B to pin C of the fuel rail pressure sensor connector. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 2M ohms	**2A**
	Replace the fuel rail pressure sensor Refer to Procedure 019-115 (/qs3/pubsys2/xml/en/procedures/96/96-019-115.html).	**3A**

STEP 2. Check the Centinel™ harness.

STEP 2A. Inspect the Centinel™ harness and Centinel™ control module connectors.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the Centinel™ harness from the Centinel™ control module.

Action	Specification/Repair	Next Step
- Loose connector - Corroded pins - Bent or broken pins - Pushed back or expanded pins - Moisture in or on the connector - Missing or damaged connector seals - Dirt or debris in or on the connector pins - Wire insulation damage - Connector shell broken - Damaged locking tab connector. For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html).	No damaged pins	**2B**
	Repair the damaged pins Repair or replace the Centinel™ harness or the Centinel™ control module, whichever has the damaged pins. - Repair the Centinel™ harness. Refer to Procedure 019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html). - Replace the Centinel™ harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html). - Replace the Centinel™ control module. Refer to Procedure 019-130 (/qs3/pubsys2/xml/en/procedures/96/96-019-130-tr.html).	**3A**

STEP 2B. Check for an open circuit.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the fuel rail pressure sensor from the Centinel™ harness.
- Disconnect the Centinel™ harness from the Centinel™ control module.

Action	Specification/Repair	Next Step
• Measure the resistance from pin 11 of the Centinel™ harness connector to pin A of the fuel rail pressure sensor, harness side. • Measure the resistance from pin 27 of the Centinel™ harness connector to pin B of the fuel rail pressure sensor, harness side. • Measure the resistance from pin 14 of the Centinel™ harness connector to pin C of the fuel rail pressure sensor, harness side. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	Less than 10 ohms	**2C**
	Repair or replace the Centinel™ harness Replace the Centinel™ harness. Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	**3A**

STEP 2C. Check for a short circuit to ground.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the fuel rail pressure sensor from the Centinel™ harness.
- Disconnect the Centinel™ harness from the Centinel™ control module.

Action	Specification/Repair	Next Step
• Measure the resistance from pins 11 and 14 of the Centinel™ harness connector to engine block ground. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 1k ohms	**2D**
	Replace the Centinel™ wiring harness Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	**3A**

STEP 2D. Check for a short circuit from pin to pin.

Conditions :

- Turn the keyswitch OFF.
- Disconnect the Centinel™ harness from the fuel rail pressure sensor.
- Disconnect the Centinel™ harness from the Centinel™ control module.

Action	Specification/Repair	Next Step

• Measure the resistance from pins 11, 14, and 27 of the Centinel™ harness connector to all other pins in the connector. Refer to the wiring diagram for connector pin identification. For general resistance measurement techniques, refer to the Resistance Measurements Using a Multimeter and Wiring Diagram, Procedure 019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html).	More than 1k ohms	**3A**
	Replace the Centinel™ wiring harness Refer to Procedure 019-131 (/qs3/pubsys2/xml/en/procedures/96/96-019-131-tr.html).	**3A**

STEP 3. Clear the fault codes.

STEP 3A. Disable the fault code.

Conditions :

- Connect all the components.
- Turn the keyswitch ON.

Action	Specification/Repair	Next Step
• Start the engine and let it idle for 1 minute. • Verify that Fault Code 119 is inactive by checking the red light.	Fault Code 119 inactive/red light off	**Complete**
	Return to troubleshooting steps or contact your local Cummins Authorized Repair Location if all the steps have been completed and checked again.	**1A**

Last Modified: 25-Feb-2004